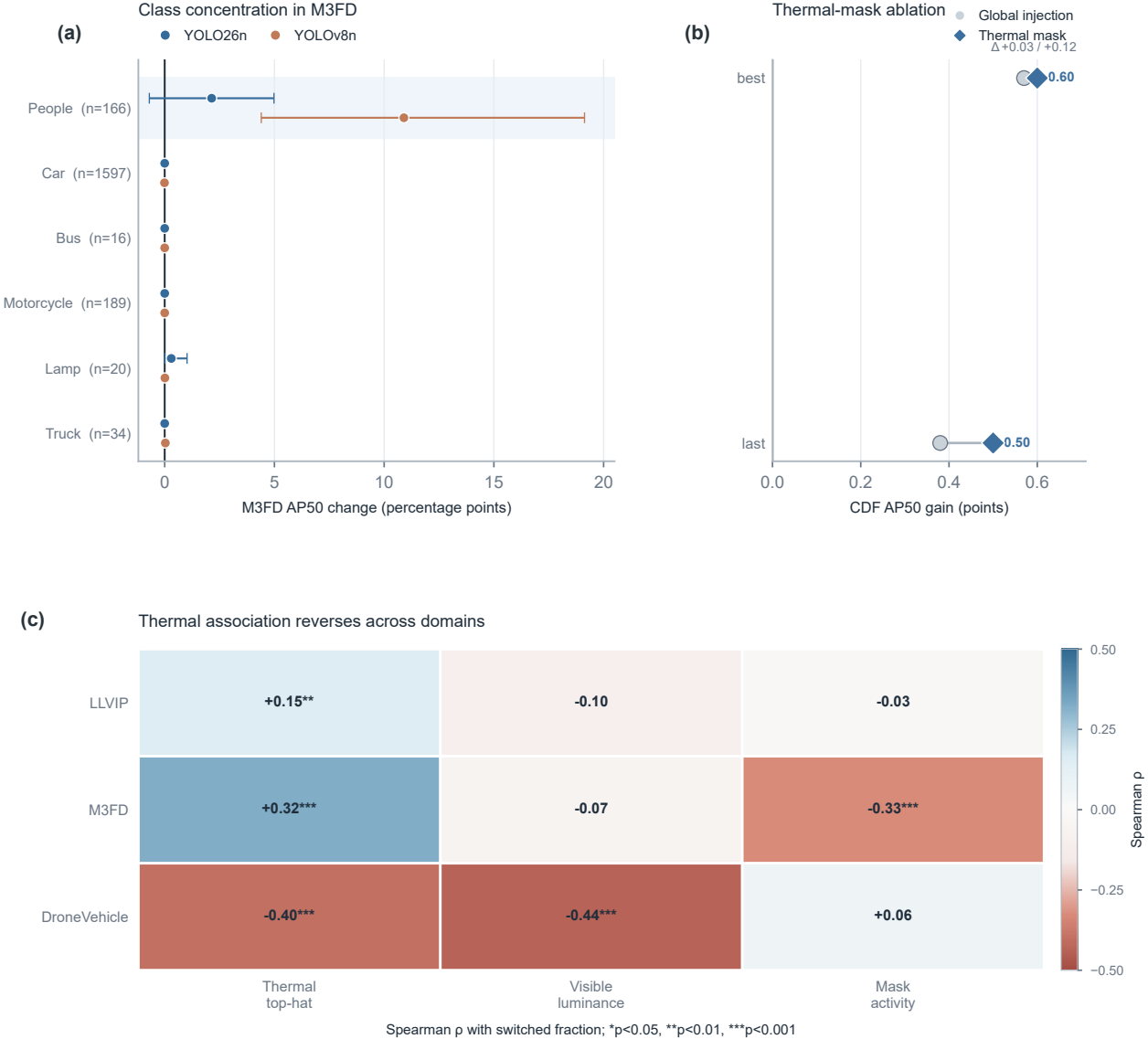

(a)
Class concentration in M3FD
YOLO26n
YOLOv8n
People (n=166)
Car (n=1597)
Bus (n=16)
Motorcycle (n=189)
Lamp (n=20)
Truck (n=34)
0
5
10
15
20
M3FD AP50 change (percentage points)
(b)
Thermal-mask ablation
Global injection
Thermal mask
Δ +0.03 / +0.12
best
last
0.60
0.50
0.0
0.2
0.4
0.6
CDF AP50 gain (points)
(c)
Thermal association reverses across domains
LLVIP
M3FD
DroneVehicle
+0.15**
-0.10
-0.03
+0.32***
-0.07
-0.33***
-0.40***
-0.44***
+0.06
Thermal top-hat
Visible luminance
Mask activity
0.50
0.25
0.00
−0.25
−0.50
Spearman ρ
Spearman ρ with switched fraction; *p<0.05, **p<0.01, ***p<0.001